Ranking

Problem 1. Suppose there are $N = 100$ travel options available for traveling from a source location to a destination. Each option P_i is characterized by two objectives: travel time T_i and travel cost C_i, both of which are to be minimized and are considered equally important.

The objective is to rank all N travel options using the concept of non-dominated sorting. An option P_i is said to dominate ($\prec$) another option P_j if

$$T_i \leq T_j, \qquad C_i \leq C_j,$$

and **at least one of these inequalities is strict**.

The objective is to rank all N travel options using non-dominated sorting. The rank of an option P_i is defined as

$$r(P_i) = \begin{cases} 1, & \text{if no option dominates } P_i, \\ 1 + \max\limits_{P_j \prec P_i} r(P_j), & \text{otherwise.} \end{cases}$$

Thus, Rank 1 contains all non-dominated options, and each subsequent rank contains the non-dominated options among those remaining after removing all preceding fronts.

Example: Consider the following six travel options, where travel time is measured in minutes and travel cost in rupees.

Option	Travel time (T_i)	Travel cost (C_i)	Rank $r(P_i)$
P_1	30	1000	1
P_2	40	700	1
P_3	50	500	1
P_4	45	900	2
P_5	60	800	3
P_6	70	1200	4

Table 1: Travel options and their non-dominated ranks.

The first front consists of P_1, P_2, and P_3, since none of these options is dominated by another option. Option P_4 is dominated by P_2 and therefore belongs to Rank 2. Option P_5 is dominated by P_4 and belongs to Rank 3. Finally, P_6 is dominated by P_5 and belongs to Rank 4.

The task is to determine the non-dominated rank of each of the 100 travel options based on their travel times and costs.